

# Assignment 1

## Q1. Python Data Types — int, float, str, bool

Python has four fundamental data types. Each stores a different kind of value:

| Type  | Description                     | Example           |
|-------|---------------------------------|-------------------|
| int   | Whole numbers, no decimal point | age = 20          |
| float | Numbers with a decimal point    | price = 99.5      |
| str   | Text enclosed in quotes         | name = "Gargi"    |
| bool  | Only True or False              | is_student = True |

### Code Example:

```
x = 10 # int
y = 3.14 # float
z = "Hello" # str
flag = True # bool

print(type(x)) # <class 'int'>
print(type(y)) # <class 'float'>
print(type(z)) # <class 'str'>
print(type(flag)) # <class 'bool'>
```

*Key difference:* int stores exact whole numbers, float stores decimals, str stores text, and bool stores only two logical values (True/False).

## Q2. Variables — name, age, city

Variables are containers for storing data. We assign values using the = operator.

```
name = "Gargi"
age = 20
city = "Jaipur"

print("Name:", name)
print("Age:", age)
print("City:", city)
```

### Output:

```
Name: Gargi
Age: 20
City: Jaipur
```

## Q3. Name Input — Uppercase & Length

input() reads a value typed by the user. upper() converts it to capitals. len() counts the number of characters.

```
name = input("Enter your name: ")

print("Uppercase:", name.upper())
print("Total characters:", len(name))
```

### Output (if input is 'Gargi'):

```
Uppercase: GARGI
Total characters: 5
```

upper() converts all letters to capitals. len() counts every character including spaces.

## Q4. Five Common String Methods

String methods are built-in functions that operate on text (str) values.

```
s = " hello world "
```

# 1. upper() - converts to uppercase  
s.upper() → " HELLO WORLD "

# 2. lower() - converts to lowercase  
s.lower() → " hello world "

# 3. strip() - removes leading/trailing spaces  
s.strip() → "hello world"

# 4. replace() - replaces a word/character  
s.replace("world", "Python") → " hello Python "

# 5. split() - splits string into a list  
s.strip().split() → ['hello', 'world']

| Method        | Purpose                             |
|---------------|-------------------------------------|
| upper()       | All letters → UPPERCASE             |
| lower()       | All letters → lowercase             |
| strip()       | Removes extra spaces from both ends |
| replace(a, b) | Swaps a with b inside the string    |
| split()       | Breaks string into a list of words  |

## Q5. List of Five Fruits

A list stores multiple items in order. Indexing starts at 0; -1 refers to the last element.

```
fruits = ["Apple", "Banana", "Mango", "Grapes", "Orange"]  
  
print("Complete list:", fruits)  
print("First element:", fruits[0]) # index 0 = first  
print("Last element:", fruits[-1]) # index -1 = last  
print("Total items:", len(fruits))
```

**Output:**

```
Complete list: ['Apple', 'Banana', 'Mango', 'Grapes', 'Orange']  
First element: Apple  
Last element: Orange  
Total items: 5
```

In Python, indexing starts at 0. Using -1 always gives the last item regardless of list size.

## Q6. List Operations — append() and remove()

append() adds a new item to the end of a list. remove() deletes an item by its value.

```
numbers = [10, 20, 30, 40, 50]  
  
numbers.append(60) # adds 60 at the end
```

```
numbers.remove(20) # removes first occurrence of 20  
  
print("Updated list:", numbers)
```

***Output:***

```
Updated list: [10, 30, 40, 50, 60]
```

append() always adds to the end. remove() deletes by value, not by position.

## Q7. What is Artificial Intelligence (AI)?

**Definition:** AI is the ability of a computer or machine to perform tasks that normally require human intelligence — such as understanding language, recognising images, making decisions, and learning from experience.

### *Importance of AI:*

- Automates repetitive tasks, saving time and reducing human error.
- Helps analyse huge amounts of data quickly and accurately.
- Enables smarter decision-making in medicine, business, and science.
- Powers personalisation in apps and services we use every day.

### *Four Real-Life Applications:*

| Application    | How AI is Used                                  |
|----------------|-------------------------------------------------|
| Healthcare     | Diagnosing diseases from X-rays/MRI scans       |
| E-commerce     | Recommending products based on browsing history |
| Transportation | Self-driving cars navigating roads safely       |
| Finance        | Detecting fraudulent transactions in real time  |

## Q8. Is it AI? — Identify and Explain

A system is considered AI if it **learns from data** and **adapts its behaviour**. A simple programmed tool (like a calculator) that never learns is NOT AI.

| Example                       | AI?   | Explanation                                                                                     |
|-------------------------------|-------|-------------------------------------------------------------------------------------------------|
| ChatGPT                       | ✓ Yes | Uses large language models to understand and generate human-like text — learns from vast data.  |
| Google Maps route prediction  | ✓ Yes | Uses ML to predict traffic, suggest fastest routes, and learn from real-time data.              |
| Calculator                    | ✗ No  | Follows fixed rules only ( $2+2=4$ always). No learning, no adaptation — just programmed logic. |
| Netflix recommendations       | ✓ Yes | Uses collaborative filtering (ML) to analyse watch history and suggest shows you will enjoy.    |
| Voice Assistants (Alexa/Siri) | ✓ Yes | Use NLP + ML to understand speech, interpret intent, and respond intelligently.                 |

**Key Rule:** If a system learns from data and adapts its behaviour, it is AI. A calculator never adapts — it just computes. That is the core difference.